

Sample Project Approach (Roadmap)

TITLE OF THE PROJECT		
STARTUP KPI DASHBOARD		
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Title: Startup KPI Dashboard for Business Performance Analytics

Domain

Business Analytics

Dataset

<https://www.kaggle.com/datasets/ara001/financial-metrics-of-startup-companies>

Abstract

Startups generate large volumes of operational, financial, customer, and marketing data. However, this information is often distributed across multiple sources, making it difficult for founders and managers to monitor overall business performance and make timely strategic decisions. Without a centralized analytical platform, identifying revenue trends, customer behaviour, profitability, and marketing effectiveness becomes challenging.

This project proposes a **Startup KPI Dashboard for Business Performance Analytics**, integrating data preprocessing, exploratory data analysis (EDA), KPI calculation, and interactive visualization into a unified business intelligence solution.

Initially, the collected datasets undergo preprocessing, including missing value handling, duplicate removal, data transformation, and feature engineering to ensure high-quality data. A comprehensive Exploratory Data Analysis (EDA) is then performed to understand business trends through descriptive statistics, revenue analysis, customer analysis, sales performance, regional analysis, profitability analysis, and marketing performance evaluation.

The dashboard calculates important business KPIs such as Total Revenue, Profit, Profit Margin, Revenue Growth Rate, Customer Acquisition Cost (CAC), Customer Retention Rate, Average Revenue Per User (ARPU), Monthly Sales Growth, and Marketing Return on Investment (ROI).

Interactive dashboards are developed using Power BI, allowing users to monitor startup performance through KPI cards, trend analysis, sales charts, customer segmentation, profit analysis, and regional performance. Business intelligence techniques enable stakeholders to identify growth opportunities, monitor financial health, optimize marketing strategies, and improve customer retention.

Finally, business insights and recommendations are generated based on the analytical findings, enabling data-driven decision-making for sustainable startup growth

Approach:

Data Analytics

Raw Dataset



Dataset Overview



Data Quality Assessment



Descriptive Statistics



Missing Values



Duplicate Records



Outlier Detection



Revenue Analysis



Sales Analysis



Customer Analysis



Marketing Performance Analysis



Profitability Analysis



Regional Analysis



Correlation Analysis



Feature Engineering



KPI Calculation



Preprocessing

- Missing Value Handling
- Duplicate Record Removal
- Data Cleaning
- Data Transformation
- Date Formatting
- Feature Engineering
- KPI Calculation

Business KPI Analysis

- Revenue Analysis
- Profit Analysis
- Sales Trend Analysis
- Customer Growth Analysis
- Customer Retention Analysis
- Customer Acquisition Cost (CAC)
- Marketing ROI Analysis
- Regional Performance Analysis

Visualization

- KPI Cards
- Revenue Trend (Line Chart)
- Sales by Product (Bar Chart)
- Customer Segmentation (Pie/Donut Chart)
- Profit Margin Analysis
- Heatmap
- Boxplot
- Interactive Dashboard (Power BI)

Comparative Analysis

Compare:

- Revenue vs Expenses
- Revenue vs Profit
- New Customers vs Returning Customers
- Product-wise Sales
- Region-wise Revenue
- Marketing Spend vs Revenue
- Monthly Performance Trends

Business Metrics

- Total Revenue
- Net Profit
- Profit Margin
- Revenue Growth Rate
- Customer Acquisition Cost (CAC)
- Customer Retention Rate
- Average Revenue Per User (ARPU)
- Return on Investment (ROI)

Recommended Research Pipeline:

Business Dataset

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Data Cleaning

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Exploratory Data Analysis (EDA)

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Feature Engineering

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KPI Calculation

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Business Performance Analysis

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Interactive Dashboard Development

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Business Insights

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Recommendations

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Project Conclusions